

TOUSSUS-LE-NOBLE AP, France

WMO: 071460

Lat: 48.7506N

Lon: 2.1133E

Elev: 164

StdP: 99.37

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.8	-3.2	-8.9	1.8	-2.1	-6.6	2.2	-1.8	14.9	9.8	12.4	8.5	2.5	30	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	31.0	20.1	28.8	19.3	26.8	18.5	21.1	28.7	20.1	26.7	19.2	25.3	3.6	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.8	23.7	17.7	13.0	22.6	16.9	12.3	21.7	61.9	28.7	58.2	26.7	55.2	25.4	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.2	9.9	8.4	-7.7	35.4	2.8	2.5	-9.6	37.3	-11.3	38.7	-12.8	40.2	-14.8	42.0	
			-8.1	23.1	2.8	1.0	-10.1	23.9	-11.7	24.5	-13.3	25.1	-15.3	25.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	4.0	4.6	7.5	10.6	14.0	17.4	19.3	18.9	15.8	12.1	7.4	4.6
	DBStd	6.48	3.83	3.75	3.47	3.45	3.34	3.35	3.30	3.10	2.93	3.46	3.42	3.75
	HDD10.0	754	187	154	92	33	5	1	0	0	1	19	92	171
	HDD18.3	2687	444	386	336	234	140	57	26	29	86	195	329	427
	CDD10.0	1256	1	2	14	49	129	222	289	275	175	83	13	3
	CDD18.3	149	0	0	0	1	6	29	57	46	11	1	0	0
	CDH23.3	1534	0	0	0	14	68	291	585	466	104	6	0	0
	CDH26.7	471	0	0	0	1	8	86	196	161	20	1	0	0
Wind	WSAvg	3.9	4.5	4.7	4.4	4.0	3.8	3.5	3.4	3.1	3.1	3.7	4.1	4.5
Precipitation	PrecAvg	672	58	49	55	49	62	54	61	51	54	60	61	58
	PrecMax	853	120	106	110	112	118	159	152	114	127	164	99	118
	PrecMin	415	13	7	2	3	8	2	14	2	8	7	10	4
	PrecStd	113	29	27	28	27	30	31	33	27	33	42	22	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	15.4	20.2	25.2	27.7	32.3	34.2	35.1	29.1	24.0	16.7	13.6
		MCWB	11.6	10.2	12.0	15.8	19.4	21.5	21.0	21.5	18.9	17.4	14.3	11.9
	2%	DB	11.8	12.4	17.6	22.4	25.2	29.0	30.9	30.9	26.4	21.0	15.2	12.4
		MCWB	10.4	9.4	11.4	14.2	17.0	19.7	19.8	20.0	17.9	16.5	13.3	10.9
	5%	DB	10.6	11.1	15.2	19.4	23.1	26.5	28.7	28.1	24.0	18.7	13.7	11.2
		MCWB	9.4	9.1	10.4	12.9	16.2	18.5	19.3	18.8	17.4	15.1	12.0	10.0
	10%	DB	9.4	10.0	13.2	17.1	20.8	24.3	26.6	25.7	21.8	17.1	12.3	10.1
		MCWB	8.3	8.3	9.8	11.7	14.9	17.6	18.5	17.8	16.1	14.3	10.9	9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	11.3	13.3	16.7	19.9	22.9	22.0	22.3	20.1	18.3	15.1	12.3
		MCDB	12.9	13.2	17.5	23.7	26.3	30.8	31.2	32.1	26.2	21.7	16.0	13.1
	2%	WB	10.6	10.2	12.2	14.9	18.3	20.7	21.0	20.8	18.9	16.7	13.5	11.2
		MCDB	11.5	11.7	15.6	21.1	23.9	26.7	28.7	28.6	24.5	19.8	14.9	12.2
	5%	WB	9.5	9.4	11.2	13.4	16.7	19.4	20.0	19.6	17.8	15.7	12.2	10.2
		MCDB	10.5	10.8	14.3	18.4	21.4	24.9	26.8	26.2	22.6	18.4	13.4	11.2
	10%	WB	8.4	8.4	10.2	12.2	15.5	18.2	19.1	18.6	16.9	14.8	11.1	9.1
		MCDB	9.3	9.9	12.9	16.1	19.7	23.4	25.4	24.4	20.8	16.7	12.3	10.1
Mean Daily Temperature Range	5% DB	MDBR	4.9	6.5	8.4	10.0	10.3	10.8	11.2	11.4	10.5	8.0	5.8	5.1
		MCDBR	5.3	8.1	12.1	13.7	13.7	14.6	15.0	15.7	13.9	9.8	6.6	5.2
		MCWBR	4.6	5.8	7.0	7.0	6.8	6.9	6.2	6.6	6.6	5.9	5.1	4.5
	5% WB	MCDBR	5.1	7.1	9.8	12.4	12.1	12.8	13.5	13.5	11.8	8.8	6.3	5.1
		MCWBR	4.7	5.7	6.3	6.8	6.7	6.8	6.1	6.4	6.1	5.7	5.1	4.6
Clear-Sky Solar Irradiance	taub	0.332	0.353	0.387	0.404	0.411	0.413	0.412	0.401	0.375	0.370	0.342	0.326	
	taud	2.389	2.342	2.241	2.213	2.232	2.263	2.261	2.311	2.372	2.373	2.394	2.361	
	Ebn at Noon	719	788	816	842	851	849	844	836	823	757	698	668	
	Edh at Noon	68	89	115	131	134	131	129	117	100	85	67	62	
All-Sky Solar Radiation	RadAvg	0.92	1.70	2.88	4.36	5.14	5.74	5.43	4.72	3.65	2.15	1.11	0.77	
	RadStd	0.10	0.24	0.36	0.49	0.49	0.54	0.44	0.45	0.30	0.17	0.11	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page